

MISSION FIELD, MT, USA

WMO: 726798

Lat: 45.698N

Lon: 110.441W

Elev: 4643

StdP: 12.39

Time Zone: -7.00 (NAM)

Period: 00-19

WBAN: 24150

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-9.1	-2.7	-15.9	2.7	-8.6	-9.3	3.9	-1.6	50.7	35.7	47.6	34.4	9.4	320	0.782

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	34.2	93.1	59.9	90.0	59.2	86.8	58.5	63.2	84.1	61.7	82.7	60.2	81.0	12.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.5	81.0	68.1	54.3	74.7	66.8	52.4	69.4	66.0	30.8	83.9	29.7	82.3	28.6	81.0	71.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 41.7	36.5	32.7	DB	-19.5	98.6	6.8	2.1	-24.4	100.1	-28.4	101.3	-32.3	102.4	-37.2	103.9	(4)
(5)			WB	-19.9	66.4	6.7	2.1	-24.8	67.9	-28.7	69.1	-32.4	70.3	-37.3	71.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.9	29.1	27.8	36.4	42.7	50.8	59.7	69.3	66.8	57.9	46.0	35.6	27.3	(6)	
	DBStd	17.63	12.34	13.35	11.81	8.77	7.99	6.94	5.20	5.74	8.22	10.42	13.20	12.95	(7)	
	HDD50	3415	648	623	433	246	89	9	0	0	23	193	448	703	(8)	
	HDD65	7286	1112	1042	886	669	443	184	20	49	239	592	882	1168	(9)	
	CDD50	1917	1	1	12	27	114	301	597	521	259	68	16	0	(10)	
	CDD65	311	0	0	0	0	2	25	153	105	25	1	0	0	(11)	
	CDH74	6379	0	0	0	10	135	641	2681	2123	742	47	0	0	(12)	
	CDH80	2722	0	0	0	0	22	219	1269	943	261	8	0	0	(13)	
(14) Wind	WSAvg	14.2	20.2	17.0	15.8	13.3	11.5	10.6	9.3	9.8	10.8	14.3	18.1	19.2	(14)	
(15) Precipitation	PrecAvg	15.4	0.6	0.4	0.8	1.4	2.6	2.6	1.3	1.5	1.7	1.5	0.8	0.4	(15)	
	PrecMax	22.9	1.1	0.9	2.2	3.2	5.2	5.3	3.2	2.7	3.1	3.5	2.4	0.7	(16)	
	PrecMin	10.7	0.1	0.0	0.3	0.5	0.6	0.6	0.2	0.2	0.0	0.2	0.1	0.0	(17)	
	PrecStd	3.3	0.3	0.3	0.5	0.8	1.1	1.3	0.9	0.9	0.9	0.8	0.7	0.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.6	57.9	68.7	75.2	82.9	91.5	97.9	96.1	90.9	79.4	66.9	54.7	(19)	
		MCWB	41.1	41.2	46.5	50.6	56.1	60.7	62.1	60.0	58.1	53.0	47.5	40.9	(20)	
	2%	DB	50.6	52.8	62.5	69.6	77.9	86.1	94.1	92.4	86.9	73.1	61.6	50.6	(21)	
		MCWB	38.2	39.5	43.8	48.8	55.1	59.7	60.6	59.2	56.6	50.7	45.2	39.1	(22)	
	5%	DB	47.7	49.2	57.5	64.5	73.3	81.3	91.0	89.1	82.8	67.9	57.4	47.3	(23)	
		MCWB	36.7	37.6	41.9	46.4	53.3	58.6	59.8	58.4	55.2	48.6	43.0	36.9	(24)	
	10%	DB	44.7	45.7	53.3	59.4	68.4	76.8	87.7	85.7	78.0	63.1	53.4	44.3	(25)	
		MCWB	34.8	35.2	40.1	44.6	51.7	57.3	59.6	57.7	53.7	46.8	40.6	35.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.1	42.9	47.4	52.8	59.8	64.3	66.1	63.9	60.5	55.0	49.4	42.2	(27)	
		MCDB	53.5	54.3	65.6	70.9	75.7	83.3	86.0	87.7	81.6	75.7	64.6	52.7	(28)	
	2%	WB	39.4	40.4	44.8	49.8	56.8	62.0	63.8	61.5	58.3	52.0	46.2	40.1	(29)	
		MCDB	48.9	51.6	60.6	66.6	72.9	80.2	84.6	83.8	80.9	69.5	60.2	49.5	(30)	
	5%	WB	37.2	38.2	42.6	47.2	54.7	60.1	62.3	60.1	56.5	49.9	43.7	37.7	(31)	
		MCDB	46.6	48.4	56.0	63.1	70.9	77.4	83.6	82.5	78.2	65.1	56.0	46.5	(32)	
	10%	WB	35.2	35.5	40.6	45.0	52.4	58.3	61.0	58.9	54.6	47.6	41.1	35.2	(33)	
		MCDB	43.8	45.0	52.3	58.4	66.7	74.8	82.5	81.0	74.8	61.4	52.4	43.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	16.7	18.5	21.0	22.7	24.7	27.5	34.2	34.2	29.9	23.0	18.1	15.8	(35)	
		MCDBR	17.5	21.7	26.5	32.9	34.8	37.5	39.8	40.8	40.2	32.0	22.1	16.2	(36)	
	5% WB	MCWBR	10.4	12.8	13.5	16.4	15.9	15.7	13.7	14.7	16.3	15.2	11.6	9.8	(37)	
		MCWBR	17.1	20.7	24.8	31.3	32.0	32.9	34.6	34.8	35.3	29.5	22.2	16.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.245	0.252	0.271	0.310	0.320	0.327	0.335	0.346	0.319	0.287	0.259	0.241	(40)	
		taud	2.428	2.418	2.483	2.415	2.471	2.496	2.498	2.432	2.501	2.560	2.537	2.469	(41)	
	Ebn at Noon		282	300	308	302	300	298	293	286	287	284	275	272	(42)	
		Edh at Noon	22	27	29	35	34	33	33	34	29	23	19	19	(43)	
(44) All-Sky Solar Radiation	RadAvg		512	821	1191	1519	1741	2082	2240	1908	1433	895	548	406	(44)	
	RadStd		45	58	79	101	170	164	104	106	121	71	51	28	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page